

BOTANY PRESENTATION

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4830

Msc botany sem II



Taxonomy and Systematic Botany

Family: Liliaceae (Lily Family)



Systematic Position



APG Classification		Bentham and Hooker Classification	
Kingdom	Plantae	Kingdom	Plantae
<i>Clade</i>	Angiosperms	Class	Monocotyle
<i>Clade</i>	Monocots	Series	Coronarieae
Order	Liliales	Order	Liliales
Family	Liliaceae	Family	Liliaceae

Family

Lilaceae (Lily family)

- Its Monocot Family and Liliaceae are fairly large family com about 250 genera and 4000 species.



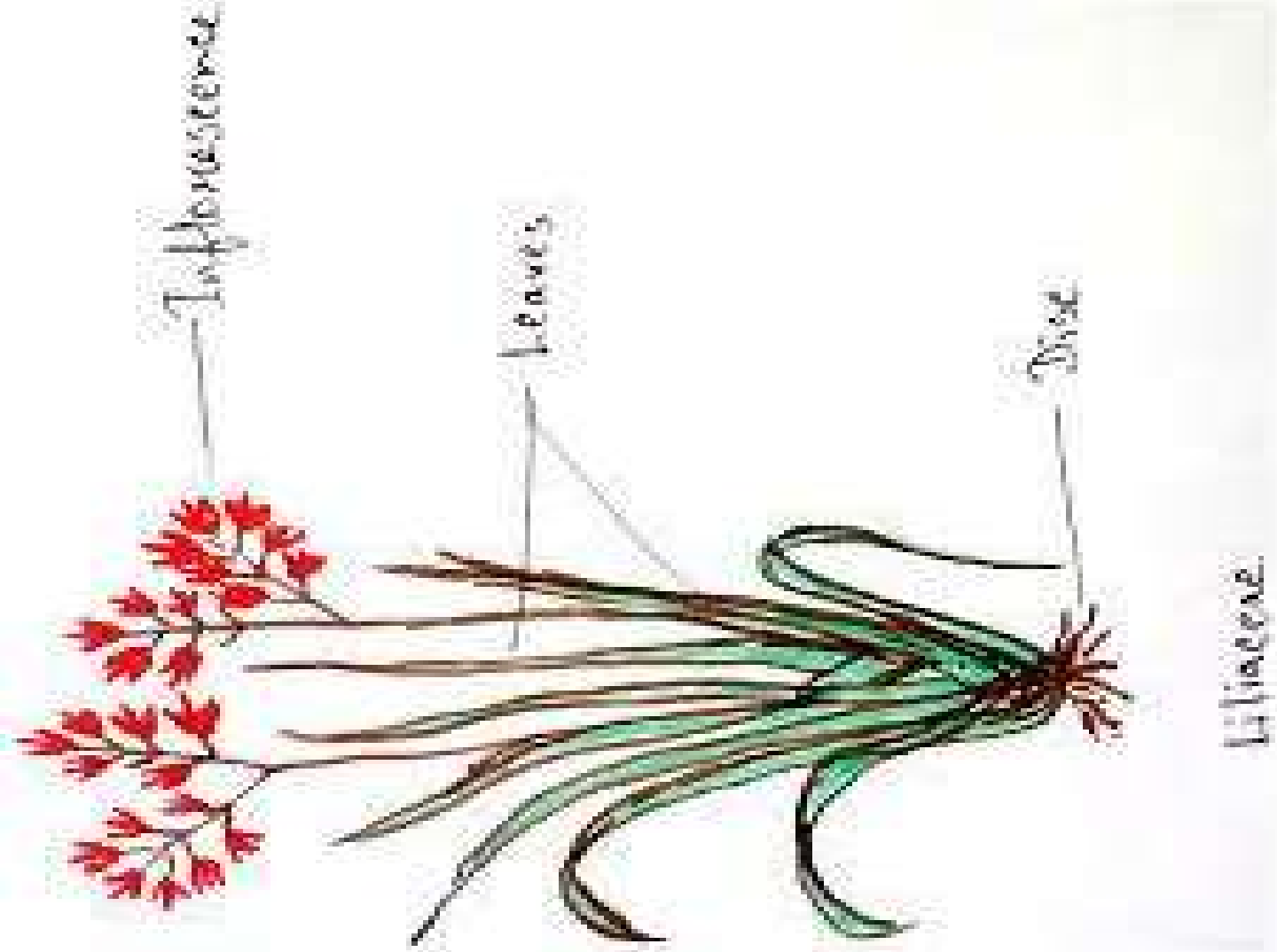
alamy stock photo



Diagnostic Features

1. Perennial herbs often with bulbous stem / rhizomes.
2. Radical leaves
3. Perianth showy.
4. Stamens six.
5. ovary superior





General Characters:-

Distribution: Liliaceae are fairly large family comprising about 7 genera and 4000 species. Members of this family are widely distributed over most part of the world.

Habit: Mostly perennial herbs persisting by means of a rhizome (*Polygonatum*), by a bulb (*Lilium*) corm (*Colchicum*), shrubby or tree like (*Yucca* and *Dracaena*). Woody climbers, climbing with the help of stipular tendrils in *Smilax*. Trees in *Xanthorrhoea*, succulents in *Al*

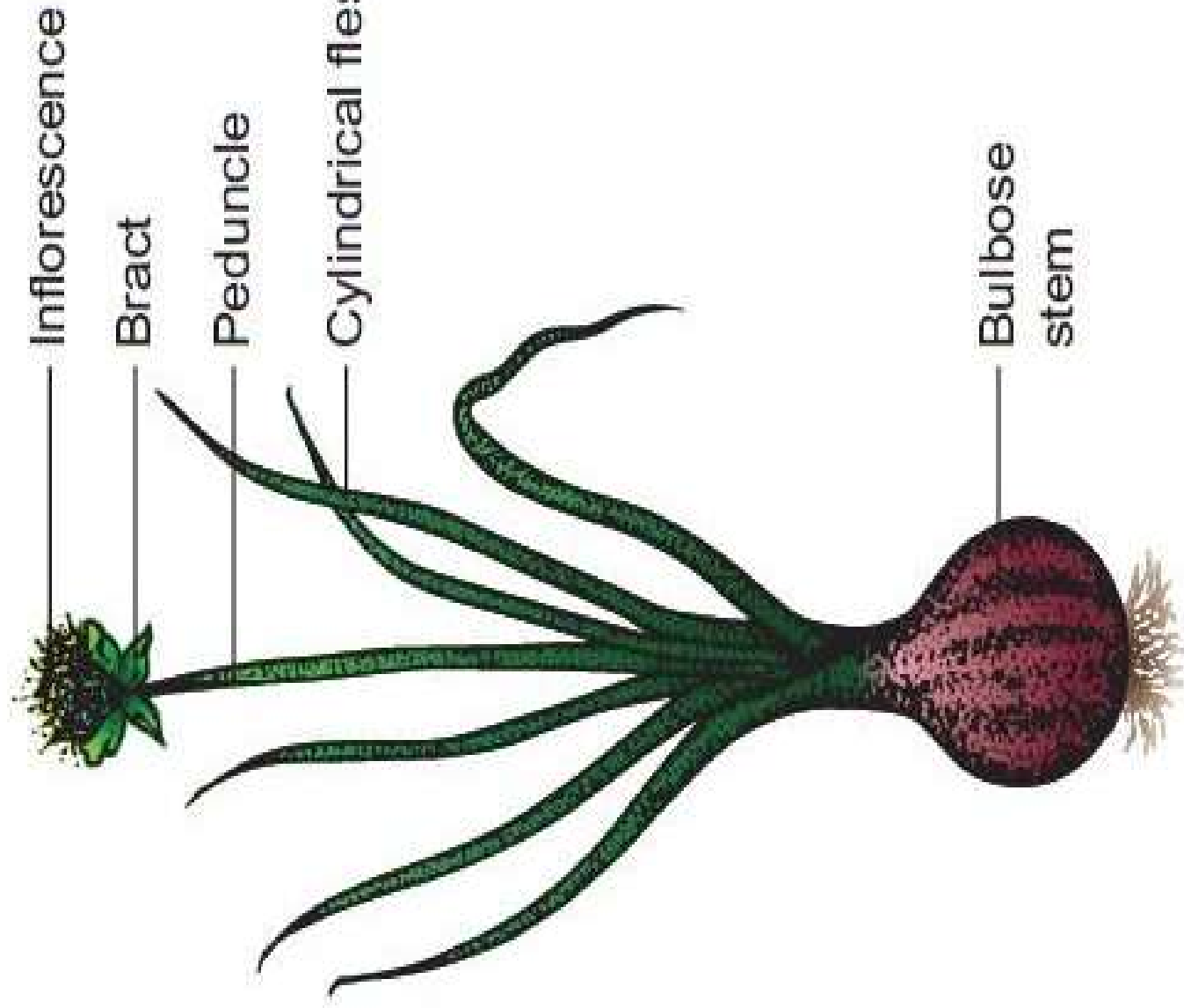


Root: Adventitious and fibrous, swollen due to storage of food

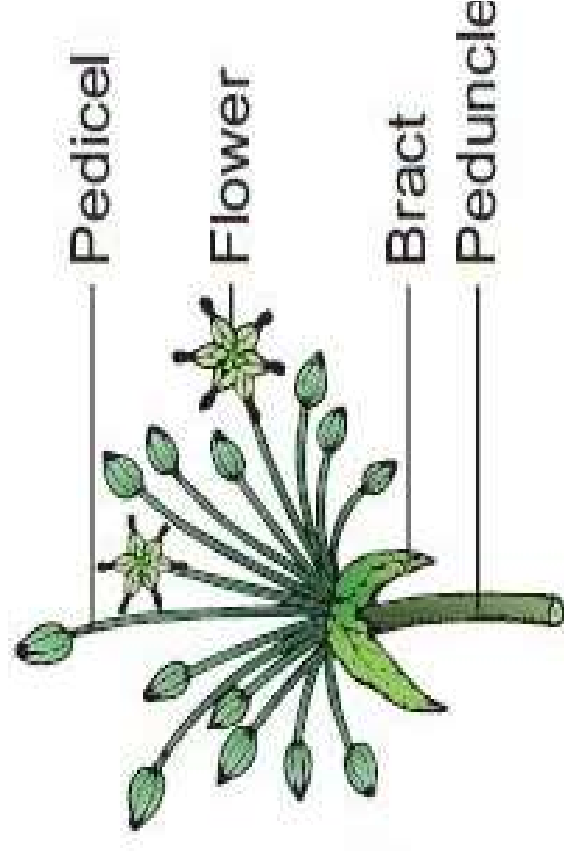
Stem: Stems usually bulbous, aerial, erect (*Dracaena*) or climbing (*Smilax*) in *Ruscus* the ultimate branches are modified into phyllodes. In *Asparagus* stem is modified into cladodes and the leaves are reduced to scales.

Leaf: Leaves are radical (*Lilium*) or cauline (*Dracaena*), usually alternate, opposite (*Gloriosa*), sometimes fleshy and hollow, reduced to scales (*Ruscus* and *Asparagus*). The venation is parallel but in species like *Smilax* it is reticulate. Leaves are usually exstipulate, but in *Smilax* tendrils arise from the base of the leaf, which are considered modified stipules.

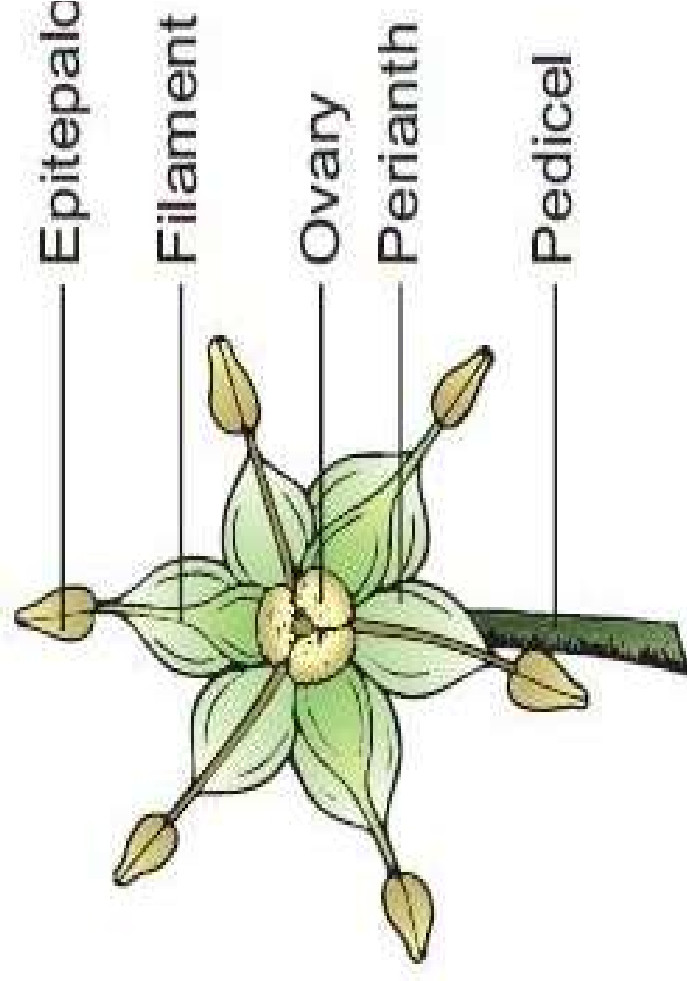




Habit



Inflorescence



Flower entire

Inflorescence:

Variable-solitary (*Tulipa*, *Fritillaria*), panicked raceme (*Asphodelu*
cymose umbel (*Allium*, *Smilax*), solitary axillary (*Gloriosa*).



Flower:

Pedicellate, actinomorphic or zygomorphic (*Lilium*, *Hemerocallis*), hermaphrodite or unisexual in *Smilax*, *Ruscus*; hypogynous, complete or incomplete (in unisexual flowers), trimerous rarely 2 or 4-merous, Bracteate

Perianth: Six, two in whorls of three each, poly or gamophyllous, odd tepal of outer whorl anterior, inferior.



Androecium: Stamens 6, arranged in two whorls of three each. Epiphyllous, apostamenous / free and opposite to tepals. Anthers dithecous, basifixed, introrse, and dehiscent longitudinally.

Gynoecium: Tricarpellary and syncarpous. Ovary superior, trilobular with two ovules in each locule on axile placentation. Style simple, slender with simple stigma.

Fruit: A loculicidal capsule or berry

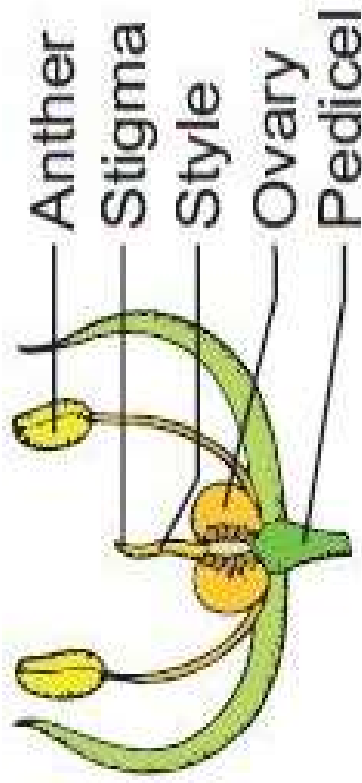
Seed: Endospermous

Floral Formula:

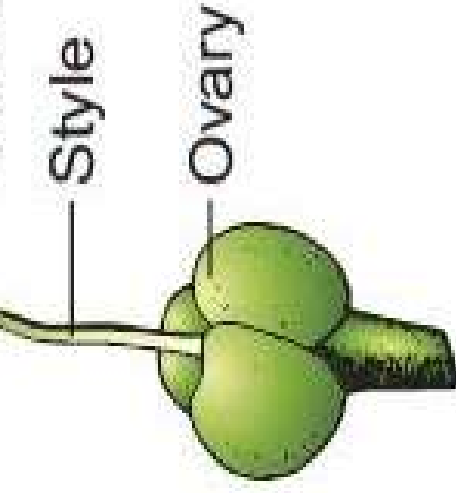
Floral Formula:

$$\text{Br.}, \text{Ebrl.}, \oplus, \overset{\uparrow}{\underset{\downarrow}{Q}}, \overset{\uparrow}{\underset{\downarrow}{P}}_{(3+3)} + \overset{\uparrow}{\underset{\downarrow}{A}}_{3+3}, \overset{\uparrow}{\underset{\downarrow}{G}}_{(3)}$$

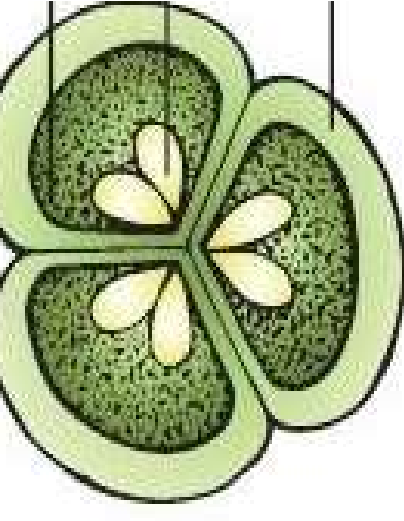




L.s of flower



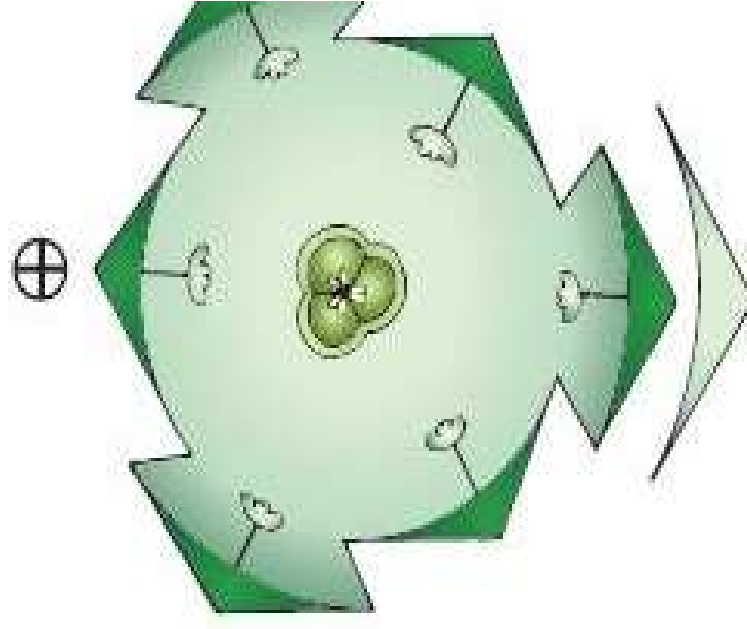
Gynoecium



C.S. of ovary

Floral formula

$$\text{Br.}, \text{Ebrl.}, \oplus, \overset{\curvearrowright}{\underset{\curvearrowleft}{Q}}, P_{(3+3)} + A_{3+3}, \underline{G}_{(3)}$$



Floral diagram

Primitive characters:

1. Perennial and arboreal habit in some genera (*Aloe*, *Yucca*).
2. Leaves simple and spirally arranged in most genera.
3. Solitary flowers in some genera.
4. Flowers hermaphrodite and actinomorphic.
5. Flowers are large and hypogynous.
6. Perianth polyphyllous in some genera (*Asphodelus*, *Veratrum*).
7. Seeds endospermic.



Advanced characters:

1. Many plants are herbaceous.
2. Leaves exstipulate, opposite or whorled in some genera.
3. Flowers are zygomorphic in some genera (*Lilium*, *Hemerocallis*).
4. Flowers unisexual (*Ruscus*, *Smilax*).
5. Perianth gamophyllous in many genera.
6. Stamens epiphyllous.
7. Reduction in the number of stamens to 3 (*Ruscus*).
8. Carpels 3 or 2 and syncarpous.
9. Axile placentation.



Common plants of the family:

1. Allium:

Cultivated in winter for bulbs.

2. Aloe:

Shrubby xerophytic plant, wild on rocks with dull red flowers.

3. Asphodelus:

Common winter weed in wheat fields.

4. Asparagus:

A thorny climbing under-shrub with cladodes and fasciculated flowers.

5. Colchicum (Meadow saffron):

Corm is very useful. It gives Colchicine.



6. Dracaena:

Tree with anomalous secondary growth in the stem.

7. Fritillaria:

Grown in garden as ornamental.

8. Gloriosa (Glory lily):

Climber with tips of leaves turned into tendrils.

9. Ruscus:

Flowers arise on the upper surface of phylloclades.

10. Smilax:

A climber with stipular tendrils and reticulate venation.

11. Tulipa:

Grown in garden as ornamentals.



Important Types of Liliaceae:

Asphodelus Tenuifolius



Allium cep



Asphodelus Tenuifolius

Habit:

An annual weed.

Root:

Adventitious, fibrous.

Stem:

Compressed, herbaceous, without nodes and internodes, smooth, greenish brown, aerial, peduncles are erect and fistular.

Leaves:

Radical, simple, cylindrical, green, pointed, smooth, apex acute, exstipulate, sessile, sheathing leaf-base, parallel venation.



Inflorescence:

Compound raceme.

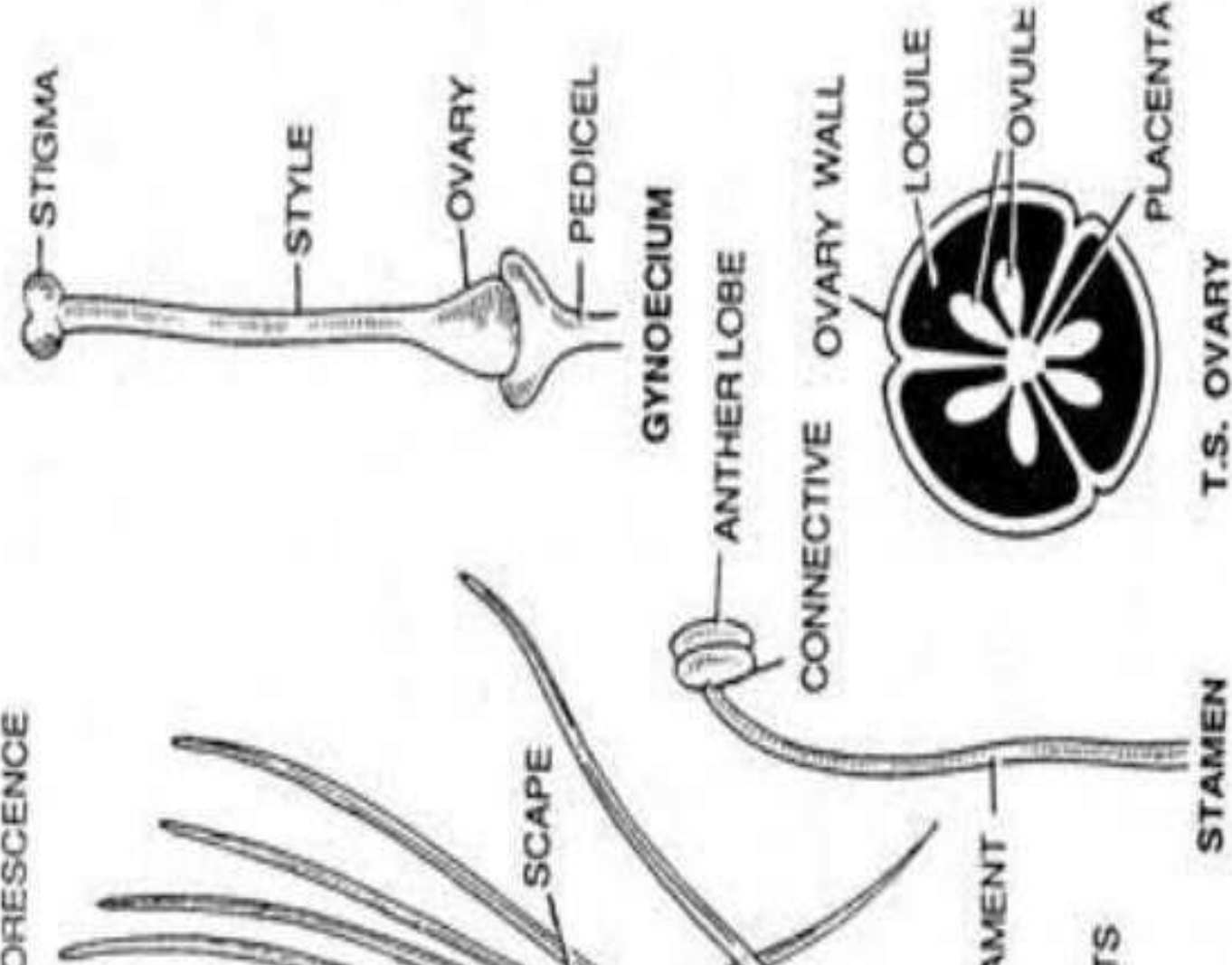
Flower:

Bracteate, pedicellate, incomplete, actinomorphic, hermaphrodite, hypogynous, trimerous, bisexual
Colour:-white

Perianth:

Tepals 6 in two whorls of 3 each, large, petaloid, polyphyllous, sometimes slightly fused at the base, imbricate aestivation inferior.





Androecium:

Stamens 6, epiphyllous, in two whorls of three each, polyandrous, anthers dithecal, dorsifixed or versatile inferior.

Gynoecium:

Tricarpellary, syncarpous, tilocular, axile placentation, 2 ovules per loculus, axile superior, style slender, filiform, stigma trilobed, yellow.

Fruit:

Loculicidal capsule.

Floral formula:

Br $\oplus$ $\times$ P₍₃₊₃₎ A₃₊

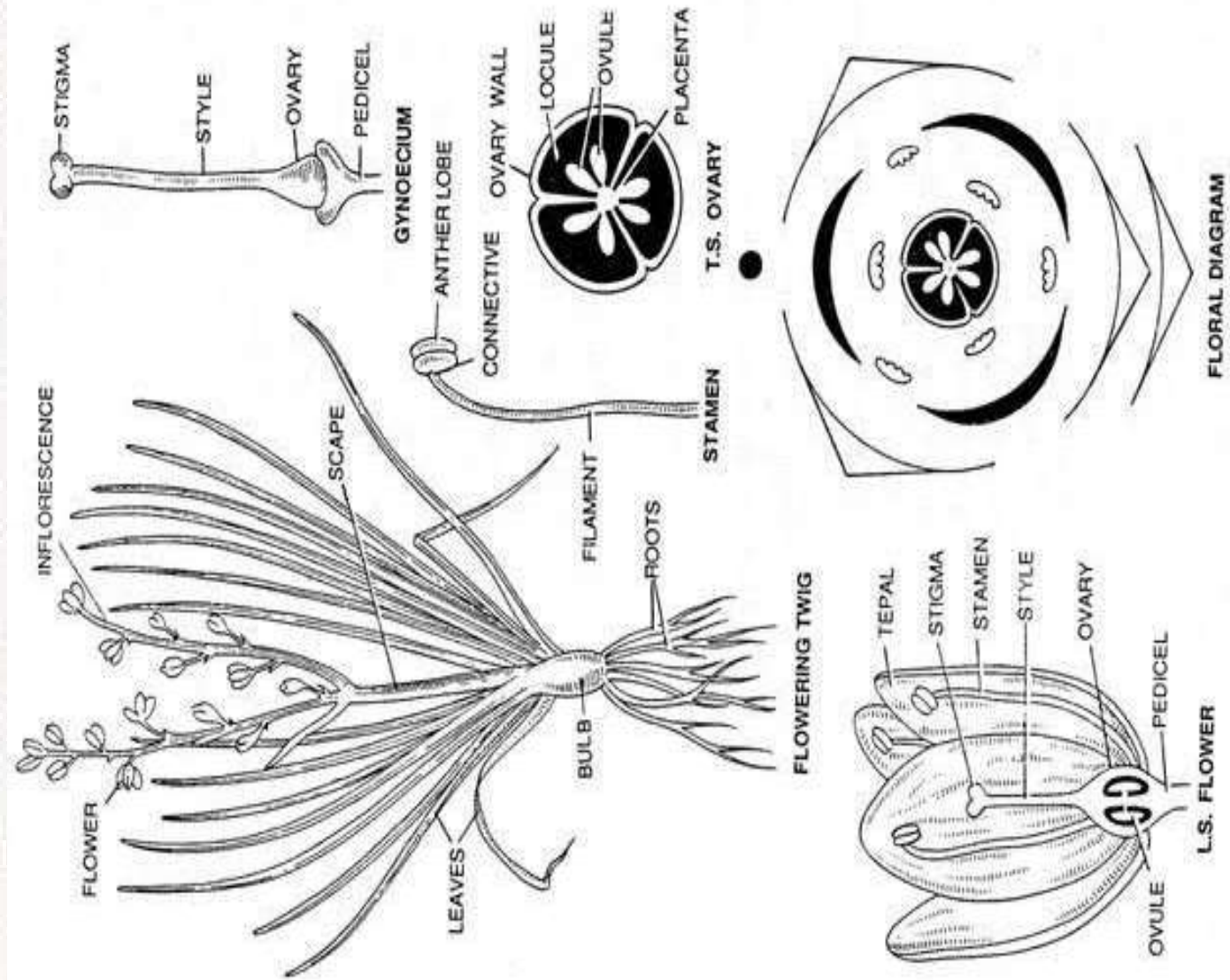
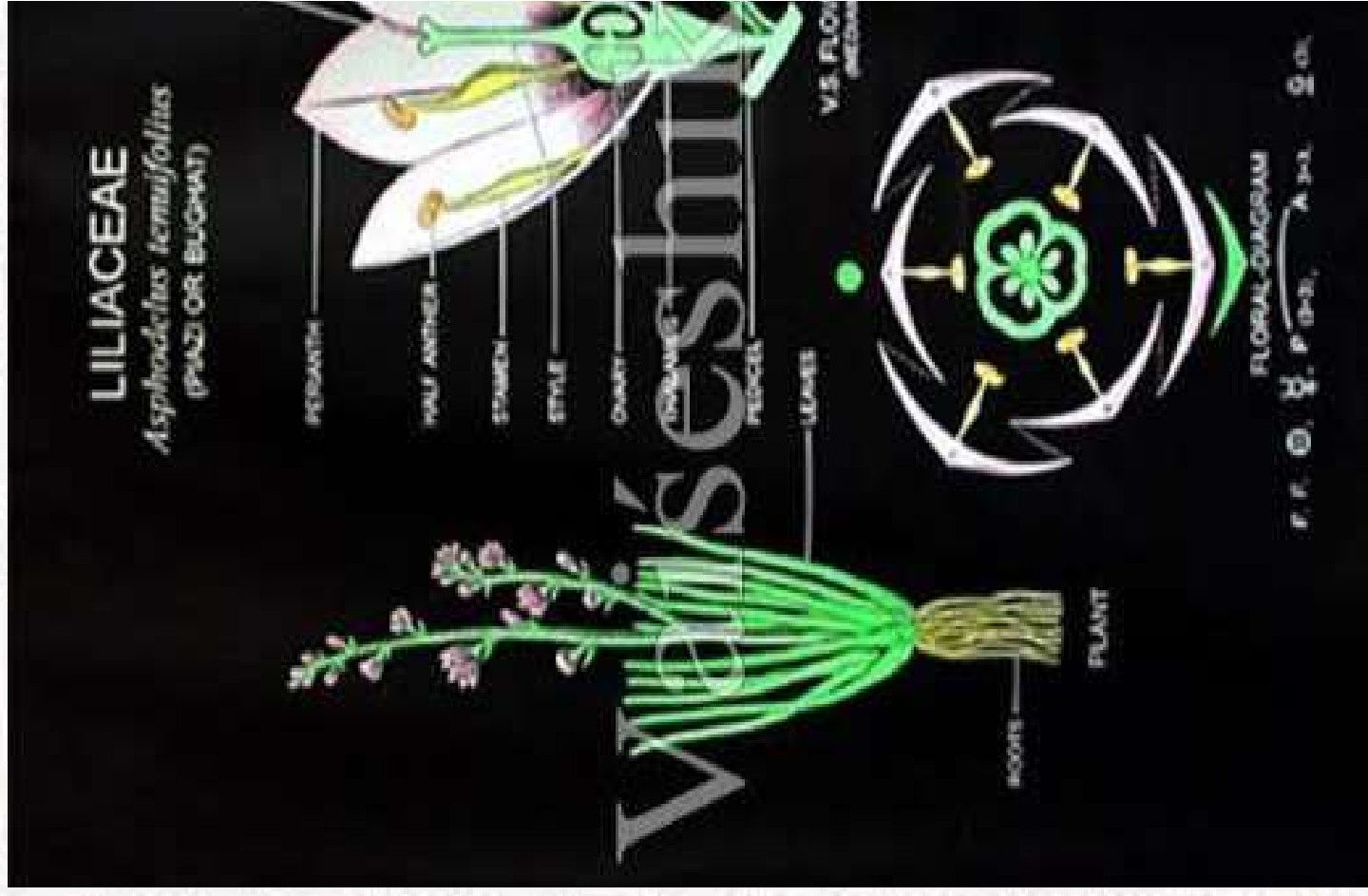


Fig. 29.1. Liliaceae. *Asphodelus tenuifolius* Cav. Verna.-Plazi.



Allium cepa

Habit:

An annual cultivated herb.

Root:

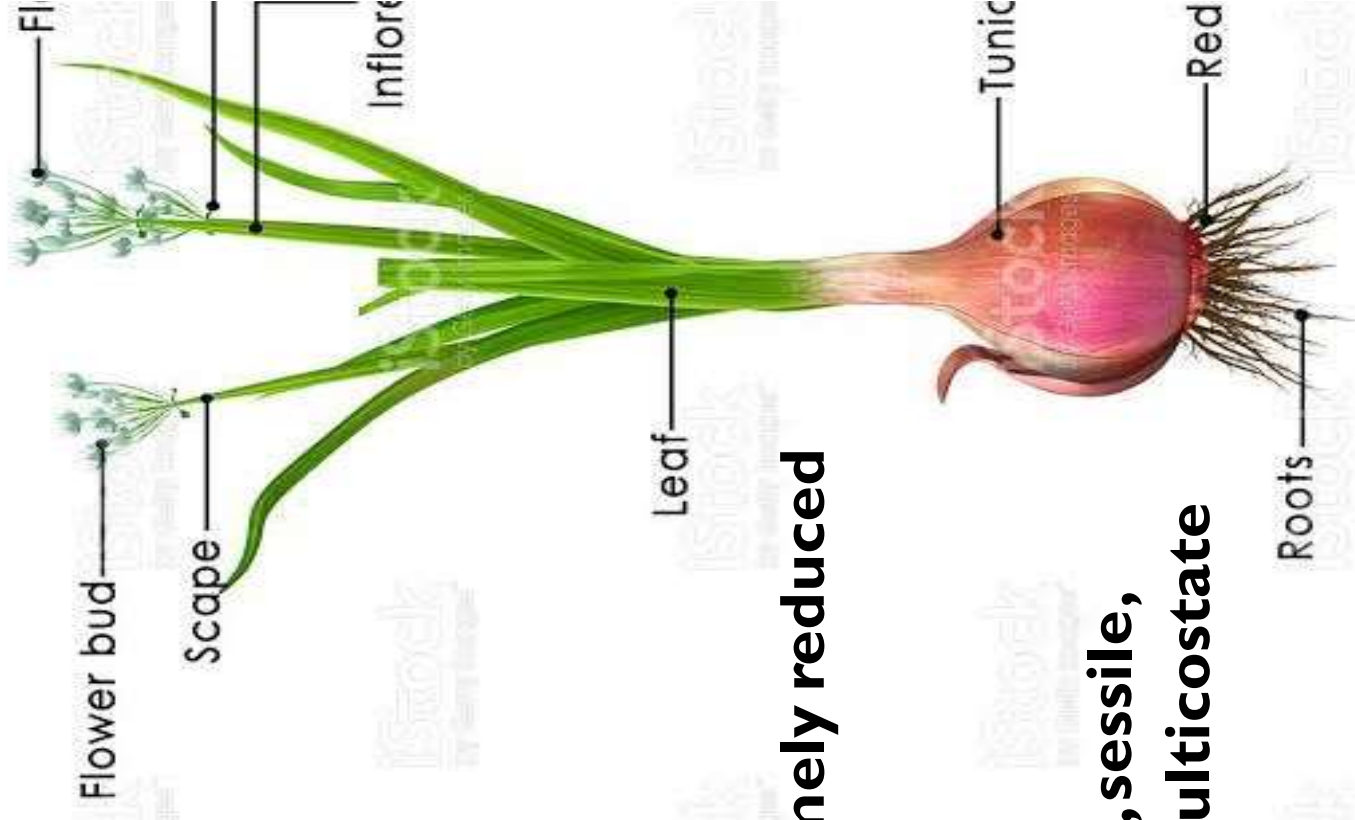
Adventitious fibrous.

Stem:

Underground tunicated bulb, extremely reduced to a conical disc like structure.

Leaves:

Radical, cylindrical, long, exstipulate, sessile, fleshy, sheathing leaf-base, hollow, multicostrate parallel venation.





Inflorescence:

Monochasial cymes enclosed by membranous bracts arranged in umbellate fashion on an erect scape.

Flower:

Pedicellate, small, hermaphrodite, hypogynous, complete, bisexual, amorphous, trimerous, white, ebracteate.



Perianth:

Tepals 6, in 2 whorls of 3 each, gamophyllous, united at the base, white, inferior.



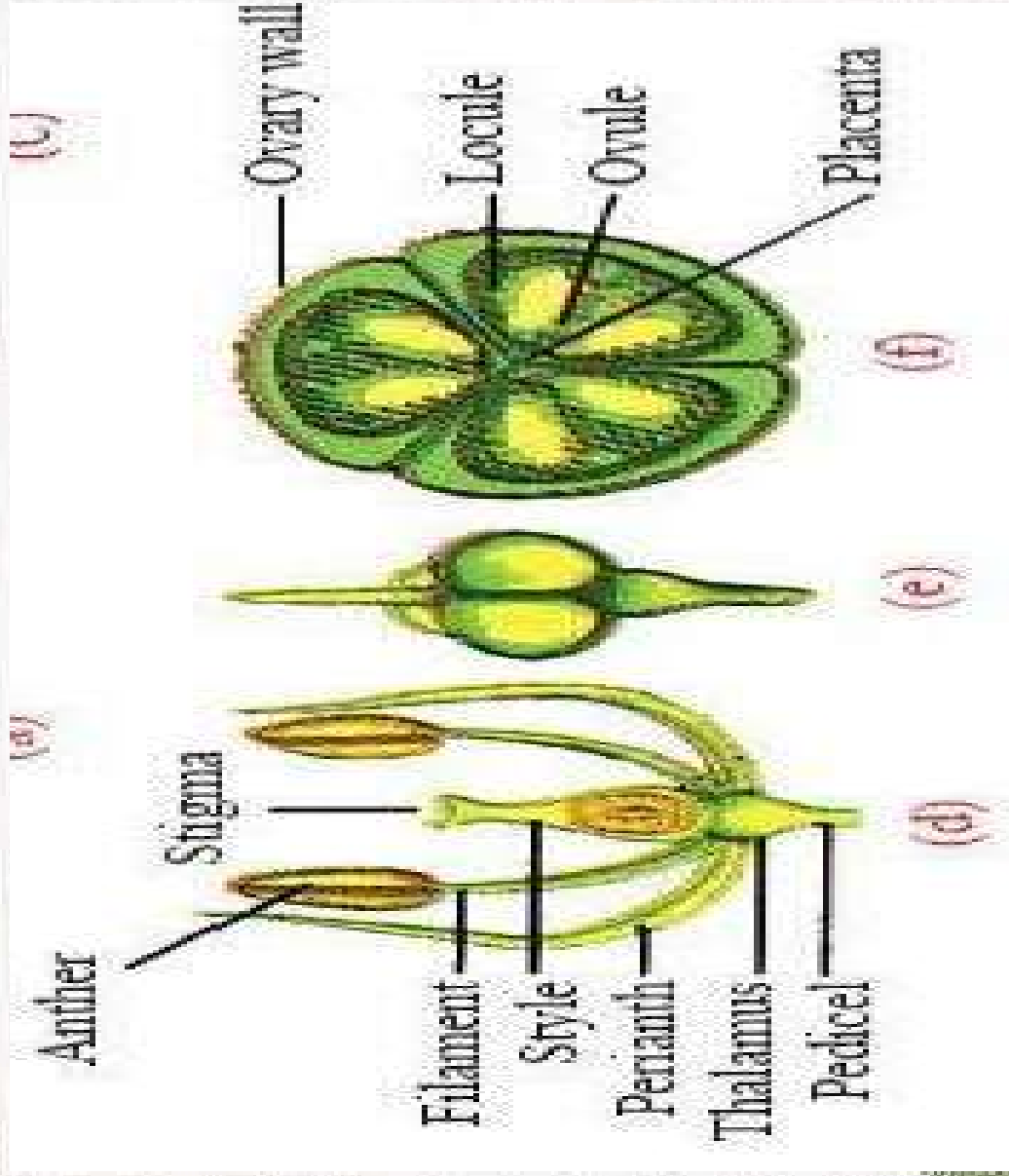
Androecium:

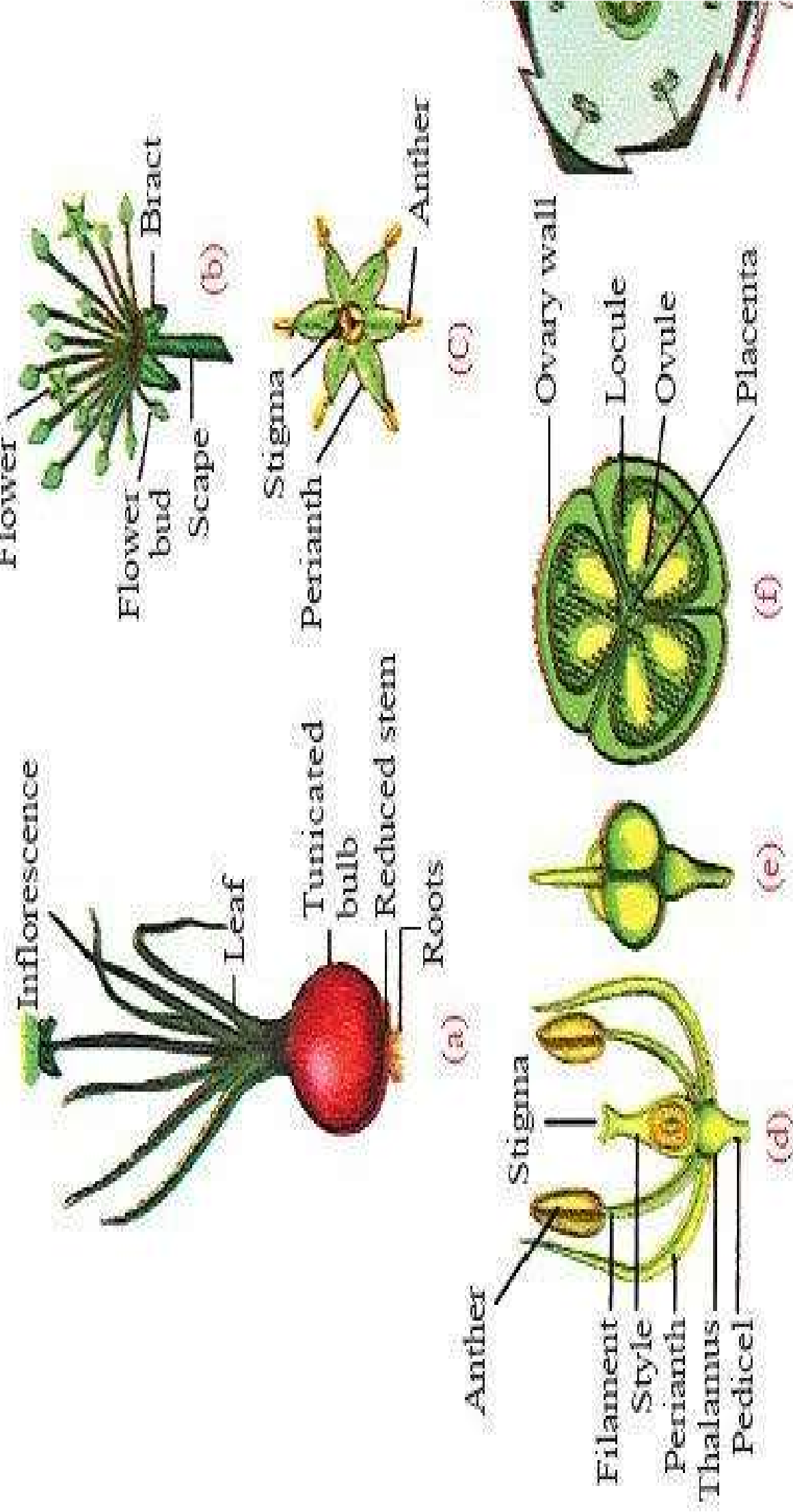
Stamens 6, polyandrous, in two whorls of 3 each, epiphyllous, filament narrow, dilated at the base, anthers dithecous, dorsifixed.

Gynoecium:

Tricarpellary, syncarpous, ovary superior, trilocular, axile placentation, 2 ovules per locule style short filiform, stigma minute.

Floral formula:





Allium cepa (onion) (a) Plant (b) Inflorescence (c) Flower (d) L.S. of flower (e) Pistil (f) T.S. of ovary (g) Floral diagram

$$\text{Floral Formula: } \text{Br } \overline{\text{Ebr}} / \oplus \overline{\text{P}}_{(3+3)} \overline{\text{A}}_3 + 3 \overline{\text{G}}_{(3)}$$

Economic importance of the family Liliaceae

- The economic importance of the plants belonging to the Liliaceae family are:
-
- Source of Medicine – *Aloe vera*, *Smilax* and *Colchicine*.
- Ornamental Plants – *Lilium*, *tulips*, *Gloriosa* and *Ruscus*.
- Source of food (or) Vegetables – *Asparagus*.
- Bulbs of *Allium cepa* and the roots of various species of *Smilax* are used as flavouring agents.



Economic Importance of Liliaceae:

1. Food

Allium cepa (Onion), *Allium sativum* (Garlic) and *Asparagus* are edible and food.

2. Medicinal:

Smilax, Aloe, Gloriosa, Veratrum, Colchicum, Scilla and Urginea yield useful Rat poison is obtained from Urginea and the bulbs of Scilla. Aloe vera yields The roots of Asparagus (H. Satavaer) yields a tonic. From Colchicum, colch obtained.

3. Fibres:

Yucca, Phormium tenax yield fibres of commerce.

4. Resin

Dracaena and Xanthorrhoea yield resin. From the acrid resin of Xantho sealing wax is prepared.

5. Ornaments

The common cultivated garden plants are Tulipa, Lilium, Gloriosa, Aloe, Dracaena, Asparagus, Yucca, Hemerocallis etc.



